A10.Author Collaborator Finder

using Graph & Network Analysis

Link Prediction on ArXiv Co-authorship Network

Course: Social Network Analysis

Dataset: ArXiv Hep-Th

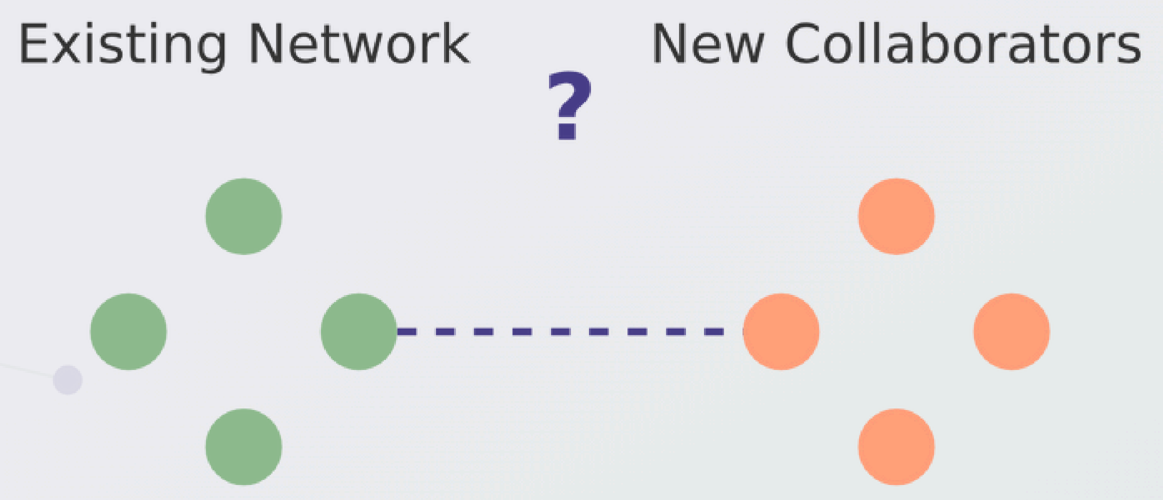
Team: 2

Nguyen Khanh Tuan - 2251607

Ha Nhat Thai - 22521316

Motivation

- **Scientific research** is highly collaborative, but discovering suitable co-authors is difficult
- **Collaboration often depends** on existing networks, creating echo chambers
- **Hard to discover** new collaborators outside one's circle
- **Large scientific datasets** enable graph-based analysis
- **Goal:** use graph structure to suggest potential future co-authors



Problem Statement

We study the link prediction problem in a co-authorship network.

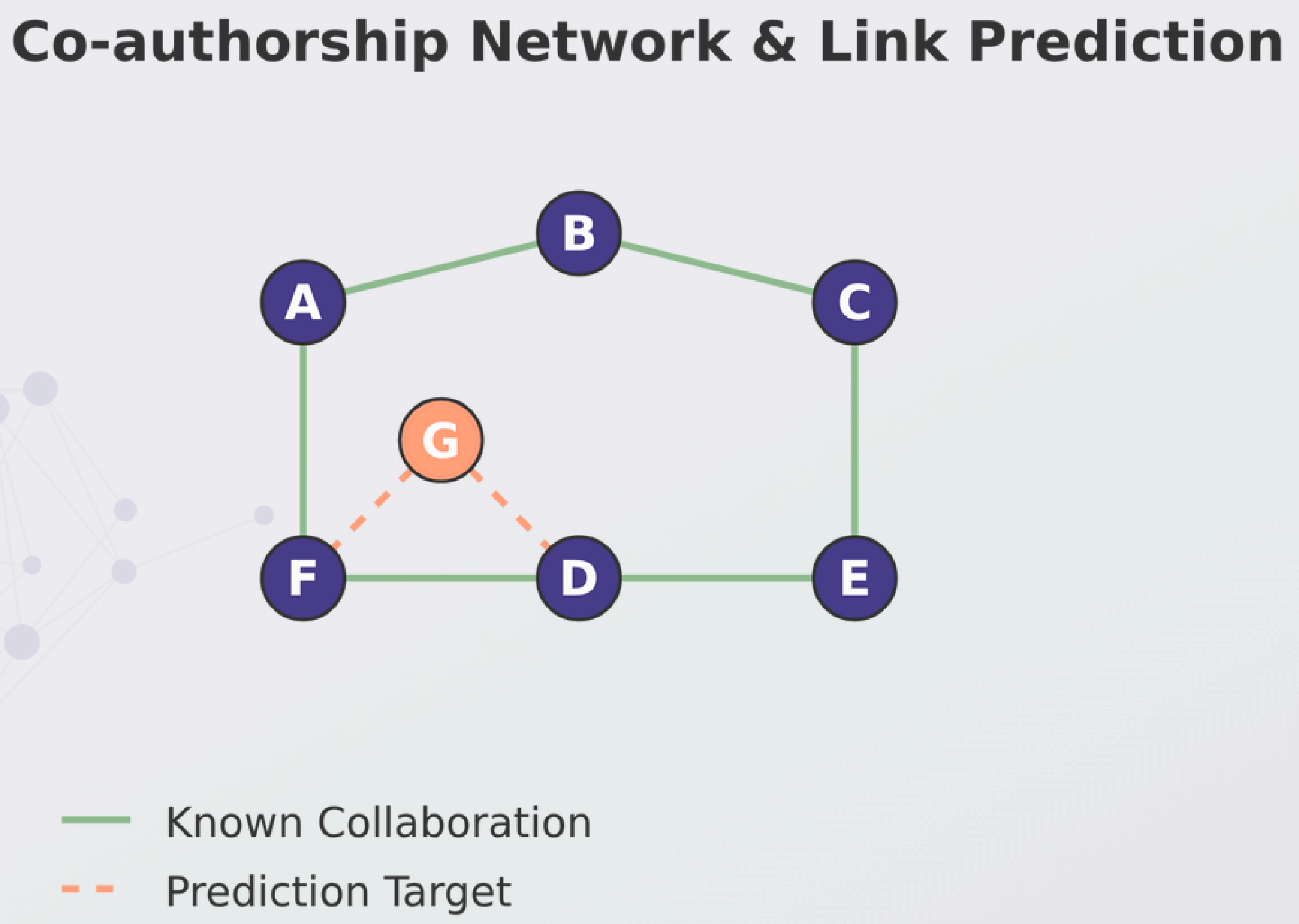
Nodes: authors

Edges: past collaborations

Task: predict future collaboration links

Research Gap

- Temporal Bias: Existing studies often use random splits, ignoring time (future leakage).
- Lack of Interpretability: Most methods are "black boxes" without clear explanations.
- Limited Scope: Often focus only on local structure or global community, rarely hybrid.



Dataset & Graph Characteristics

ArXiv Hep-Th Dataset

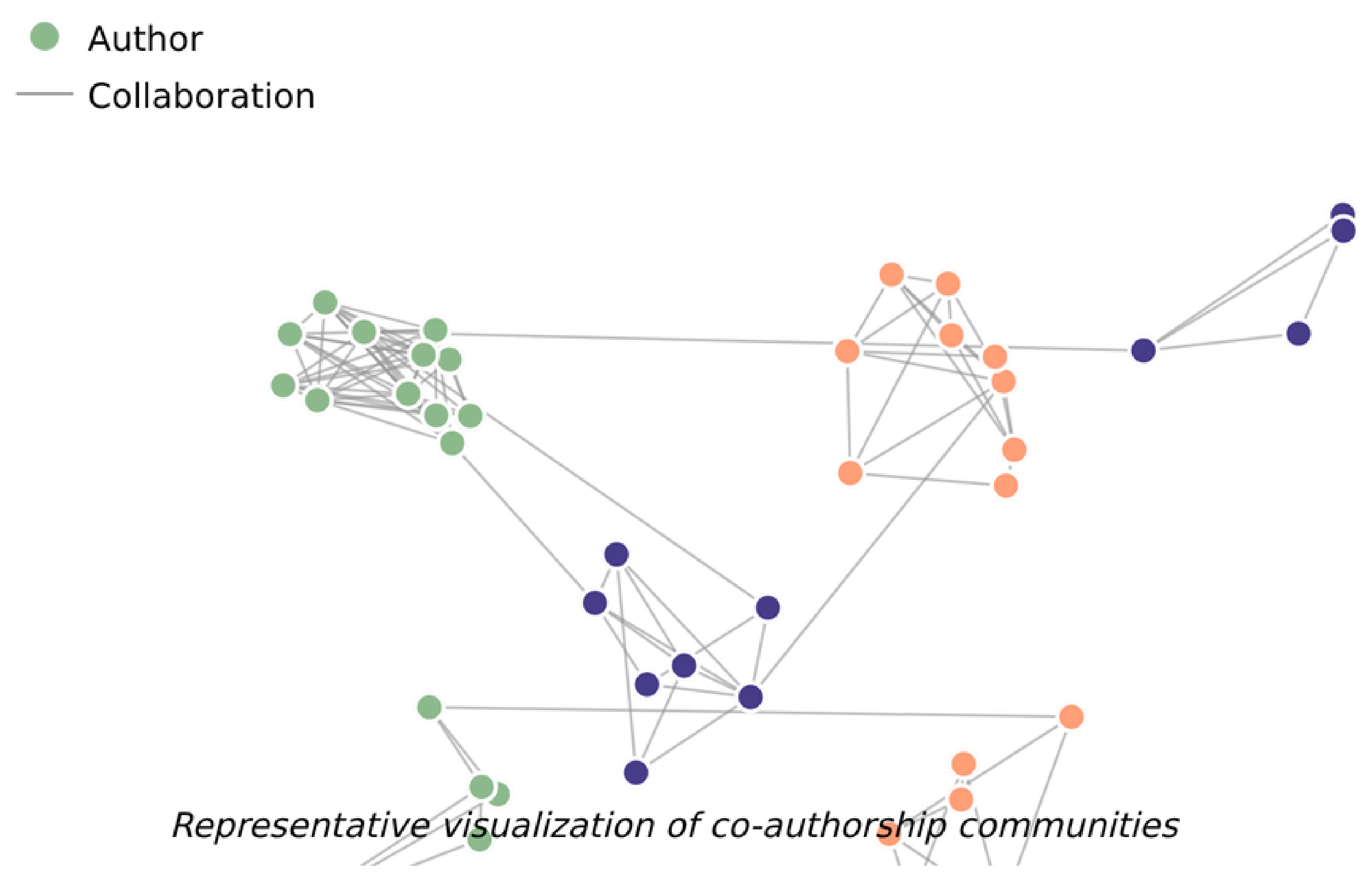
Authors
~13,003

Collaborations
~23,847

Graph Properties

- Sparse graph with strong community structure
- High clustering coefficient within communities
- Many disconnected components
- Structure motivates community-aware prediction

Network Visualization



Representative visualization of co-authorship communities

Key Insight: The graph's structure with distinct communities and sparse connections suggests that community detection will be valuable for co-author prediction.

Overall Approach Pipeline

1. Graph Construction

Build co-authorship network from data

2. Network Analysis

Study graph properties and community structure

3. Link Prediction

Apply prediction algorithms to find potential links

4. Evaluation

Measure performance with AUC and AP metrics

5. Recommendation

Generate and explain co-author suggestions

6. Prototype Demo

Demonstrate practical application of the system

Our graph-based approach enables discovery of potential co-authors beyond existing networks

Methods Overview

Local (2-hop proximity)

- Common Neighbors (CN)
- Adamic–Adar (AA)
- Resource Allocation (RA)
- Preferential Attachment (PA)

Global

- Community detection (Louvain)

Captures research communities at a global level

Hybrid

- CN + Community overlap
- AA + Community overlap
- RA + Community overlap
- PA + Community overlap

Combines local similarity with community structure

Why Multiple 2-hop Methods?

All 2-hop methods use common neighbors, but **weight them differently**. Using multiple methods helps understand which structural signal matters most.

Common Neighbors (CN)

Assumes that authors with many mutual friends are more likely to collaborate.

Adamic–Adar (AA)

Gives higher weight to shared neighbors who are less popular (low degree)

Resource Allocation (RA)

Similar to AA but penalizes popular common neighbors even more heavily

Preferential Attachment (PA)

Assumes that popular authors ("hubs") are more likely to form new connections.

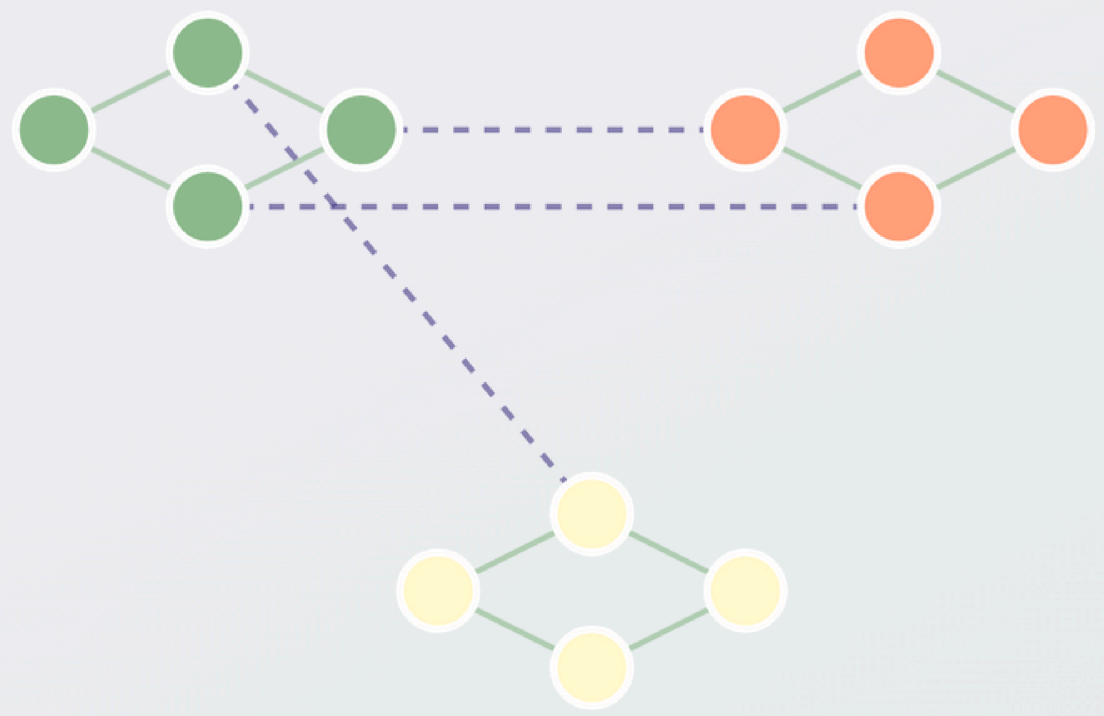
Key Insight

Different weighting strategies capture different aspects of network structure. Comparing methods helps identify which structural signals are most important for predicting collaborations.

Community Detection (Louvain)

The Louvain Algorithm

- **Detects densely connected author groups** representing research communities
- **Authors in same community** are more likely to collaborate
- **Measured by modularity** to optimize community structure
- **Adds global structural context** for improved prediction



Why It Matters

Community detection helps identify research communities, revealing hidden collaboration patterns that would be missed by local proximity measures alone.

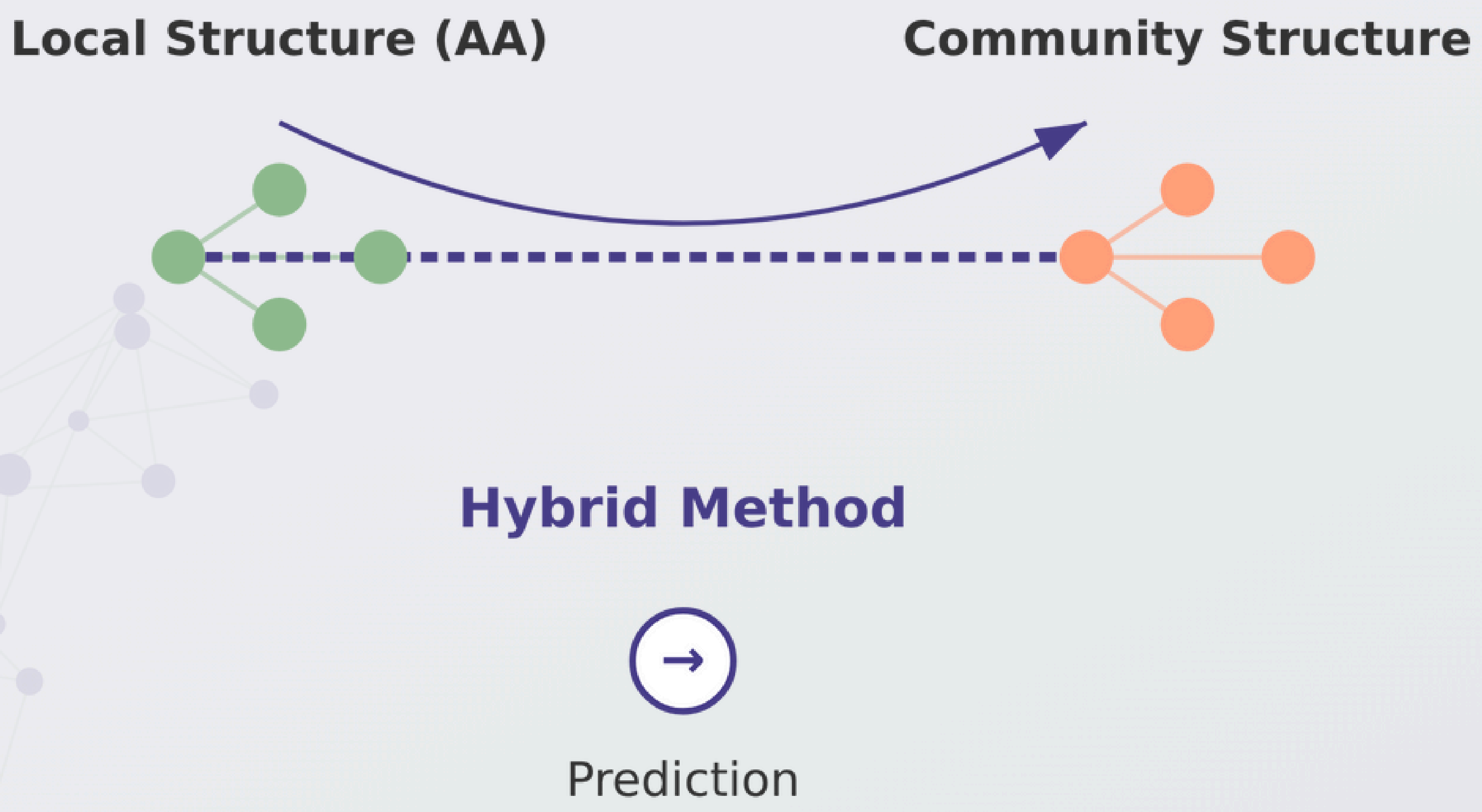
Key Insight

Authors in the same community share similar research interests and are more likely to collaborate, making community information a valuable feature for link prediction.

Hybrid Method

Our hybrid approach combines:

- **Adamic–Adar (AA)** captures fine-grained local similarity
- **Community overlap** captures global structure
- **Hybrid improves robustness** over single methods

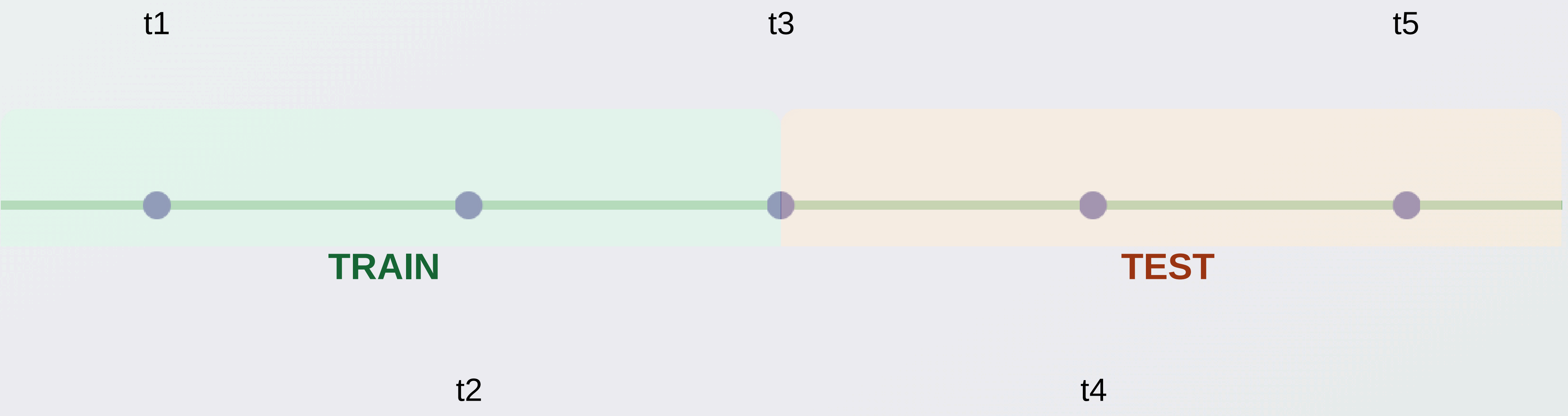


Why it works: The hybrid approach leverages both local network structure (proximity) and global community structure, providing a more complete picture of potential collaborations.

Evaluation Protocol

Protocol

- **Time-aware train/test split**: Links are ordered by time, with earlier links in training set
- **Predict future links only**: Model predicts potential collaborations that haven't happened yet
- **Balanced samples**: Equal positive and negative samples to avoid class imbalance



Metrics

AUC
Global discrimination ability: how well the model ranks positive vs. negative links across all recommendations

Average Precision (AP)
Ranking quality for Top-K recommendation: measures relevance of recommended authors

Quantitative Result

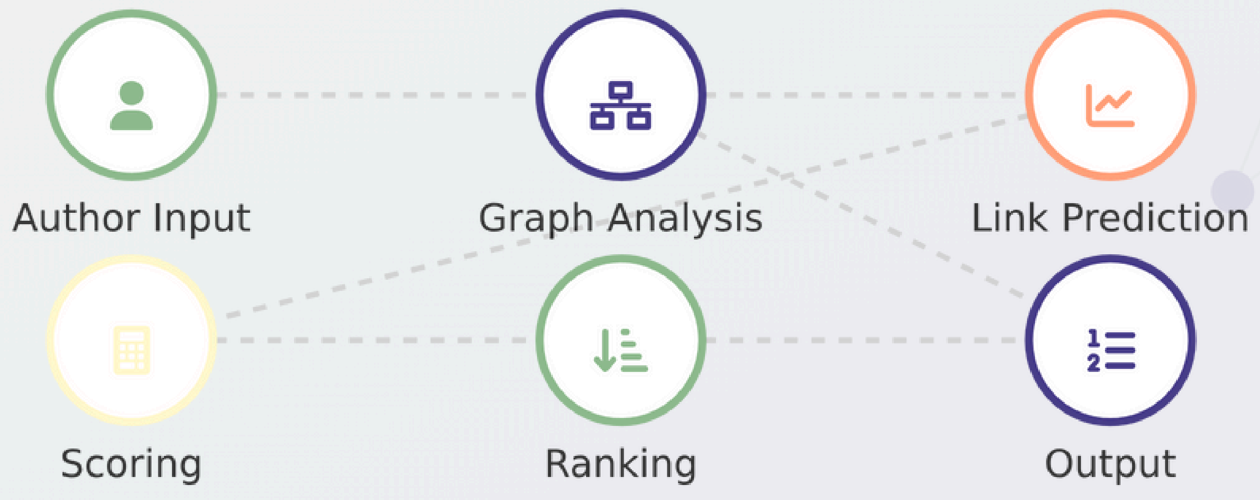
Method	AUC	AP
Common Neighbors (CN)	0.5210	0.5205
Adamic-Adar (AA)	0.5210	0.5207
Resource Allocation (RA)	0.5210	0.5206
Preferential Attachment (PA)	0.1946	0.5000
Louvain	0.5257	0.5214
Hybrid CN (HYB_CN)	0.5301	0.5296
Hybrid AA (HYB_AA)	0.5301	0.5296
Hybrid RA (HYB_RA)	0.5301	0.5295
Hybrid PA (HYB_PA)	0.1966	0.5023

Key Findings

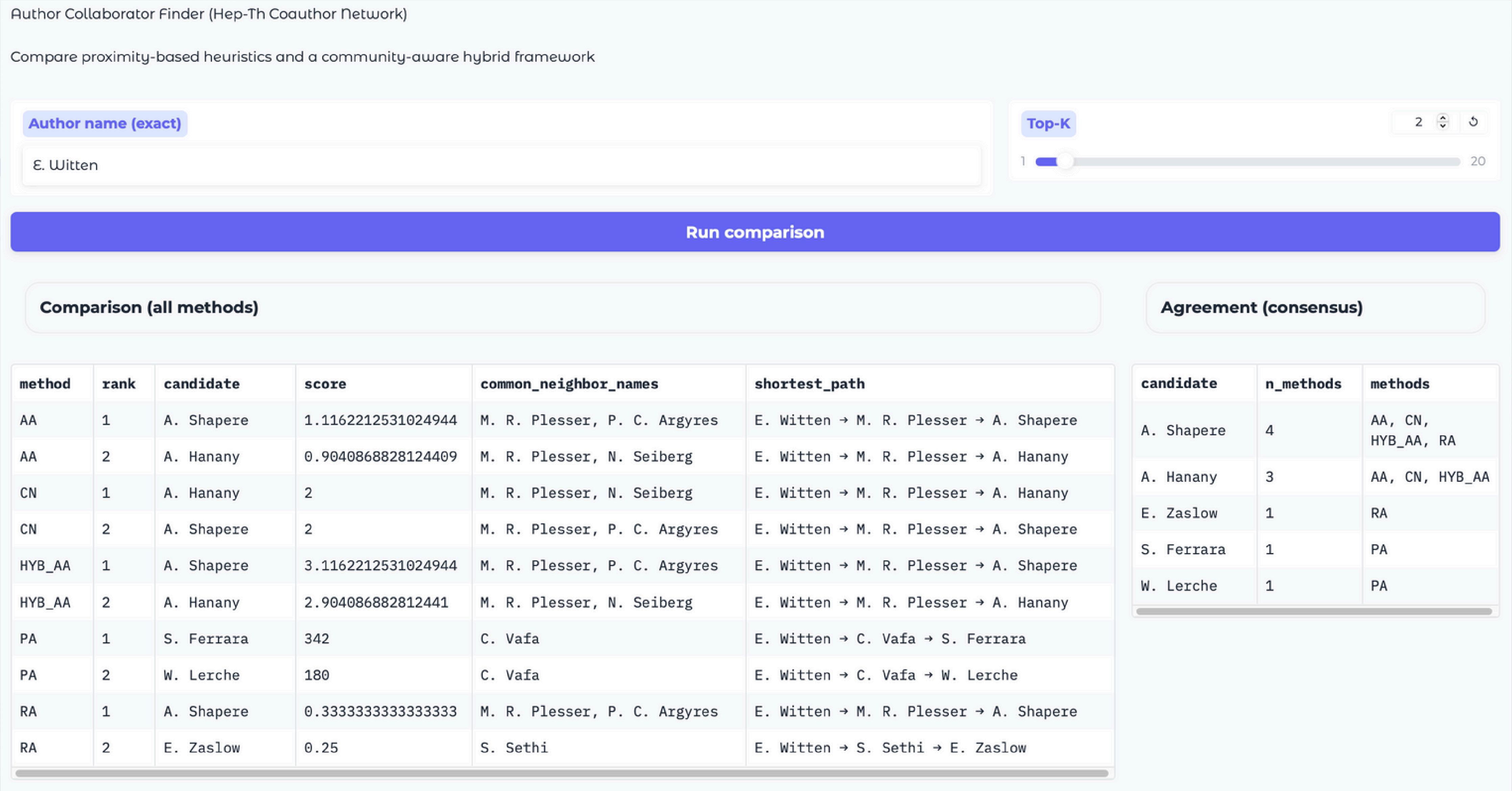
- **Preferential Attachment performs** poorly (Anti-preferential trend)
- **Community-based methods** outperform pure local proximity
- **Hybrid method** consistently improves over AA alone
- This confirms the importance of **local topology and community structure**

Prototype System

System Workflow



Interface



Author Collaborator Finder (Hep-Th Coauthor Network)

Compare proximity-based heuristics and a community-aware hybrid framework

Author name (exact)

E. Witten

Top-K

2

1 20

Run comparison

Comparison (all methods)

method	rank	candidate	score	common_neighbor_names	shortest_path
AA	1	A. Shapere	1.1162212531024944	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
AA	2	A. Hanany	0.9040868828124409	M. R. Plesser, N. Seiberg	E. Witten → M. R. Plesser → A. Hanany
CN	1	A. Hanany	2	M. R. Plesser, N. Seiberg	E. Witten → M. R. Plesser → A. Hanany
CN	2	A. Shapere	2	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
HYB_AA	1	A. Shapere	3.1162212531024944	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
HYB_AA	2	A. Hanany	2.904086882812441	M. R. Plesser, N. Seiberg	E. Witten → M. R. Plesser → A. Hanany
PA	1	S. Ferrara	342	C. Vafa	E. Witten → C. Vafa → S. Ferrara
PA	2	W. Lerche	180	C. Vafa	E. Witten → C. Vafa → W. Lerche
RA	1	A. Shapere	0.3333333333333333	M. R. Plesser, P. C. Argyres	E. Witten → M. R. Plesser → A. Shapere
RA	2	E. Zaslow	0.25	S. Sethi	E. Witten → S. Sethi → E. Zaslow

Agreement (consensus)

candidate	n_methods	methods
A. Shapere	4	AA, CN, HYB_AA, RA
A. Hanany	3	AA, CN, HYB_AA
E. Zaslow	1	RA
S. Ferrara	1	PA
W. Lerche	1	PA

Top-K recommendations

Scoring system

Justification paths

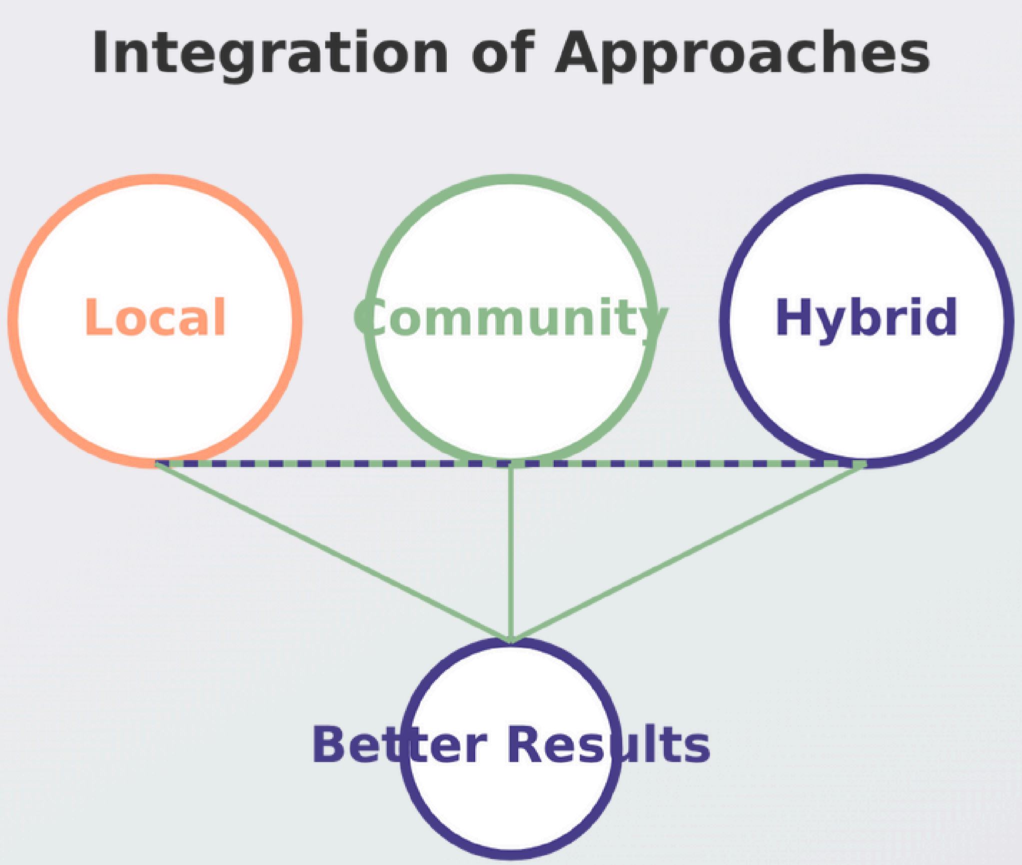
Conclusion

Successfully built a co-author recommendation system that bridges theory and application

Compared multiple link prediction methods including local, community, and hybrid approaches

Community and hybrid approaches consistently improved over single methods

Prototype demonstrates practical usability of graph-based co-author recommendation



Future Work

Semantic Features

- Add topic modeling
- Include abstract content

Temporal Modeling

- Temporal Decay Integration link prediction
- Collaboration trend analysis

Graph Neural Networks

- Explore deep graph embeddings
- Implement GNN-based models

Multi-domain Evaluation

- Test on larger datasets
- Apply to other scientific domains

Advancing research through collaborative innovation

DEMO

https://colab.research.google.com/drive/1PXnFZByrrCca89eJ8fzlrJnMNKovRjIC?usp=sharing

THANKS!

https://github.com/nhatthaiuit/SocialNetworkAnalysis.git